

CHE 312: Transport Phenomena II: Heat and Mass

Course Syllabus – Fall 2018

Class meetings:	Monday, Wednesday, Friday 1:00 pm - 1:50 pm	CEB 214
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Instructor:

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Office hours: TBA; or by appointment

Teaching Assistant:

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Objectives and Prerequisites

Introduce students to heat and mass transport phenomena. Specific topics include heat conduction, convection, and radiation, heat exchanger design, diffusion, and mass transfer coefficients.

Prerequisite: CHE 311, Differential Equations

Student Learning Outcomes

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Required Text

Bird, R.B., Stewart, W.E., and Lightfoot, E.N., 2007. *Transport Phenomena*, Revised 2nd Edition, Wiley and Sons. ISBN-13: 978-0470115398

Grading policy

The course grade will be based on the following:

In-term exams (3)—45%
Assignments and Pop Quizzes—20%
Final Exam—35% (Monday, Dec. 10th, 1:00p-3:00p)

Attendance policy

1. Attendance at class meetings is highly recommended.
2. If you miss a class meeting, you are responsible for the course material covered and compliance with any announcements made. Most handouts will be posted online.
3. Attendance at examination periods is mandatory. If you miss a quiz or examination, you will receive a zero for that quiz/examination. Makeup quizzes/exams will **ONLY** be given, in the case of an extreme emergency. I will make all decisions on this matter on a case-by-case basis.

Assigned Work

In this course, all assigned work, unless otherwise stated, should be completed and handed in individually. Evidence of direct copying will be addressed in accordance with the University's Academic Integrity policies (see below). Required assignments will be due at 4 pm on the due date. You are asked to prepare all work in a professional manner. Late work assignments will **NOT** be accepted.

Solutions for homework problems will be available for student review, and will be posted on Black Board after the assignment due date. Students are encouraged to **first** compare homework and the correct solution to find the source of errors in their calculations. If there is still some confusion, the student is encouraged to meet with the TA or myself. Office hours provide an ideal opportunity to discuss such matters.

Re-grade Policy: There will not be any exceptions to the following policy. If you would like an assignment (i.e., homework, quiz, or exam) re-graded, you must submit a typed explanation of your dispute along with the original graded assignment to me within one week of the *return date*. The *return date* is defined as the date that I handed the graded assignment back to students in class. This policy applies even if you were NOT in class and did not receive the graded assignment. Your entire assignment will be re-graded, and as a result your grade may increase, decrease, or stay the same.

Exam Policies

There will be three in-term exams during the semester. Any exam absences will receive a grade of **ZERO**. All exams may be cumulative, although they will emphasize the material covered since the last exam.

All cell phones must be stowed away from your chair/desk/person. The following actions are prohibited during a quiz or exam. (1) Using a cell phone (or PDA, etc.) in any manner or having it in your hand or on/under your desk. (2) Opening a browser window and/or accessing the Internet. (3) Talking with another student or passing notes.

If a student misses the final exam, they will receive a grade of **ZERO**. A make-up final exam will not be given. The student feedback assessment survey, to be conducted during the 15th week of the semester, will count for 3 points (out of 100 points in total) on the final exam.

Academic Integrity

This course and its associated coursework are being administered under the policies of the University of Illinois at Chicago (UIC) Student Code of Conduct. In keeping with UIC policy, **evidence of academic dishonesty may result in a failing grade for the course and disciplinary review by the University**. Academic dishonesty includes, but is not limited to, cheating (giving or receiving aid), fabrication/falsification, plagiarism (including not accurately referencing source material), bribes, favors or threats, examination by proxy (taking an exam for someone else), grade tampering and submitting non-original works authored by someone other than the student. Refer to the student handbook or Student Disciplinary Procedures (www.uic.edu/depts/sja) for definitions and details. For a complete review of UIC policies regarding intellectual integrity and academic honesty, please visit: <http://www.uic.edu/depts/dos/studentconduct.html>.

Course Topics

Heat Transfer:

- Fourier's Law of Heat Conduction (Section 9.1)
- Thermal Conductivity (Sections 9.2, 9.5)
- Shell Energy Balances and Temperature Distributions in Solids (Sections 10.1-10.3, 10.6-10.7)
- Unsteady Heat Conduction in Solids (Section 12.1)
- Heat Transfer in Fluids (Sections 9.7-9.8, 10.4, 10.5, 10.8)
- Equations of Change for Nonisothermal Systems (Sections 11.1, 11.2, 11.4)
- Free Convection (Sections 10.9, 11.3, 11.4-5)
- Dimensional Analysis (Section 11.5)
- Temperature Distributions in Turbulent Flow (Ch. 5, Sections 13.1-13.3)
- Heat Transfer Coefficients (Sections 14.1-14.7)
- Macroscopic Balances for Nonisothermal Systems (Sections 15.1-15.4)

Mass Transfer:

- Diffusivity and Mass Transport (Sections 17.1, 20.1)
- Concentration Distributions in Solids and Laminar Flow (Sections 18.1-18.7)
- Equations of Change for Multicomponent Systems (Sections 19.1-19.5)
- Mass Transport Coefficients (Sections 22.1-22.2)
- Macroscopic Balances for Multicomponent Systems (Sections 23.1-23.2)

Important Dates

In-term Exams (Subject to Change): Oct. 3rd, Nov. 2nd, Nov. 30th

Final Exam: Monday, Dec. 10th, 1:00p-3:00p

AIChE Code of Ethics

Members of the American Institute of Chemical Engineers shall uphold and advance the integrity, honor and dignity of the engineering profession by:

- Being honest and impartial and serving with fidelity their employers, their clients, and the public;
- Striving to increase the competence and prestige of the engineering profession;
- Using their knowledge and skill for the enhancement of human welfare.

To achieve these goals, members shall:

1. Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties.
2. Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.
3. Accept responsibility for their actions, seek and heed critical review of their work and offer objective criticism of the work of others.
4. Issue statements or present information only in an objective and truthful manner.
5. Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality.
6. Treat fairly and respectfully all colleagues and co-workers, recognizing their unique contributions and capabilities.
7. Perform professional services only in areas of their competence.
8. Build their professional reputations on the merits of their services.
9. Continue their professional development throughout their careers, and provide opportunities for the professional development of those under their supervision.
10. Never tolerate harassment.
11. Conduct themselves in a fair, honorable and respectful manner.